

HAM RADIO REFERENCE

Ham Radio General Exam Study Notes

Mnemonic cram sheet — formulas, propagation, components, band plan

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Quick facts

- **Frequency:** 300 / meter = MHz, or 300 / MHz = meters.
- **Dipole antenna:** $\frac{1}{2}$ -wave dipole = 468 / Frequency.
- **Silicon** — seven letters → diode forward drop $\approx$ **0.7 V**.
- **Germanium** → diode forward drop $\approx$ **0.3 V** (lower drop than silicon).
- **NAND** and **ZERO** both have four letters.
- **QRQ** = **Q**uicker; **QRS** = **S**lower; **QRV** = **R**ecei**V**e.
- Cap**AC**itors pass AC; in**Du**Ctors pass DC.
- **F**ifteen amp fuse for **F**ourteen gauge wire.
- **T**wenty amp fuse for **T**welve gauge wire.
- **AC frequency increases:** coil springs higher (reactance increases); capacitor holds back (reactance decreases).
- **Detectors:** product detector → **SSB/CW**; discriminator → **FM**; envelope/diode detector → **AM**.
- **Heterodyne receiver:** mixer. **Balanced modulator** + mixer: filter.
- **20 m data band:** 14.070–14.100 MHz.
- **LC oscillator:** tank circuit.
- **CW bandwidth** = 150 Hz; **SSB bandwidth** = 2,300 Hz; **FM bandwidth** = ± 5 kHz or ± 15 kHz.

Ohm's law and power

- **Ohm's law:** $E / I \cdot R$
- **Power law:** $P / E \cdot I$

Symbol	Quantity
E	Voltage in volts
I	Current in amps
R	Resistance in ohms
P	Power in watts

Series / parallel behavior:

Component	Series	Parallel
Resistor	Add	Less
Inductor	Add	Less
Capacitor	Less	Add

Measuring units:

Unit	Measures
Amp	Current
Farad	Capacitance
Henry	Inductance
Hertz	Frequency
Ohm	Resistance
Watt	Power
Volt	Voltage

ITU frequency band designations

Designation	Name	Frequency	Wavelength
ELF	Extremely low frequency	3–30 Hz	100,000–10,000 km
SLF	Superlow frequency	30–300 Hz	10,000–1,000 km
ULF	Ultralow frequency	300–3000 Hz	1,000–100 km
VLF	Very low frequency	3–30 kHz	100–10 km
LF	Low frequency	30–300 kHz	10–1 km
MF	Medium frequency	300–3000 kHz	1 km–100 m
HF	High frequency	3–30 MHz	100–10 m
VHF	Very high frequency	30–300 MHz	10–1 m
UHF	Ultrahigh frequency	300–3000 MHz	1 m–10 cm
SHF	Superhigh frequency	3–30 GHz	10–1 cm
EHF	Extremely high frequency	30–300 GHz	1 cm–1 mm

Power, modes, and operating

- **Maximum 1500 W PEP**, except **200 W PEP on 30 m**, and **100 W for beacons**.
- **RTTY / data** near center of CW allocation but LSB (170 Hz shift for amateur RTTY).
 - 80 m data: 3570–3600 kHz; 20 m RTTY: 14.070–14.100 MHz; 20 m PSK31: 14.070 MHz.
- **HF data bandwidth:** No symbol-rate (baud) limit on HF. Data is limited to a **2.8 kHz maximum bandwidth** (FCC 23-93).
- **Use General privileges immediately with a CSCE:** add "/AG" to a CW call and "Slant AG" on phone.

Minimum channel separation: CW 150–500 Hz; SSB 3 kHz.

Power multipliers:

- 20 dB = 100-fold change
- One S-unit = 6 dB = 4-fold change
- 3 dB = 2-fold change
- 1 dB loss = -20.5%

Data modes: PSK31 uses varicode; RTTY uses 5-bit Baudot with 170 Hz shift.

Sideband operation: below 14 MHz use lower sideband (LSB); above 14 MHz use upper sideband (USB).

Propagation

- **A-index:** long-term geomagnetic stability.
- **K-index:** short-term geomagnetic stability.
- **Solar flux:** radio energy at 10.7 cm.
- **D layer** absorbs; **E layer** max single hop ~1200 miles; **F2 layer** max single hop ~2500 miles.
- Low frequencies good at night (no D layer); higher HF works during day (good F-layer charge).
- Max range on a frequency just below the MUF; short range just above the LUF.
- UV and X-rays take 8 minutes to arrive; coronal mass ejections (CMEs) take 20–40 hours.
- A two-tone linearity test uses two **non-harmonically related** audio tones.

Q-signals and prosigns

Signal	Meaning	Signal	Meaning
QRP	Low power operation, ~5 W on HF	QRV	Ready to receive
QRS	Send more slowly	KN	Listening for specific station(s)
QRQ	Send faster	CL	Closing station
QSL	Acknowledge receipt	AR	End of message

Components and formulas

Wire sizes: 15 amp circuit requires 14 gauge wire; 20 amp circuit requires 12 gauge wire.

Series / parallel formulas:

	Series	Parallel
Resistors / inductors	Add — $R_1 + R_2 + R_3 \dots$	Divide — $1 / (1/R_1 + 1/R_2 + 1/R_3 \dots)$
Capacitors	Divide — $1 / (1/C_1 + 1/C_2 + 1/C_3 \dots)$	Add — $C_1 + C_2 + C_3 \dots$

Reactance: increases with frequency in a coil; decreases with frequency in a capacitor.

Impedance matching: matching impedance between source and load ensures **maximum power transfer / highest efficiency**.

Connectors:

Connector	Use	Connector	Use
DE-9	Serial data	PL-259	RF — good to ~150 MHz
DIN	Multiconductor audio/control	N	Waterproof RF — good to ~10 GHz
RCA	Audio	SMA	Small RF — good to several GHz

Component symbols to know: field-effect transistor, Zener diode, NPN transistor, transformer, tapped inductor.

Peak Envelope Power:

- $PEP = [(0.707 \cdot PEV)(0.707 \cdot PEV)] / R_L$ (PEV = peak voltage; R_L = resistive load)
- Peak-to-peak voltage = $2 \times$ peak voltage
- $RMS = 0.707 \times$ peak voltage

Ohm's law / power memory circles: cover the value you need and divide or multiply the rest:

- E = Voltage in volts; I = Current in amperes; R = Resistance in ohms; P = Power in watts.
- Examples: $P = I \times E$; $P = I^2 \times R$; $P = E^2 / R$.

Logic gates:

- **AND gate:** output is high only when **both** inputs are high.
 - **NOR gate:** output is low when **either or both** inputs are high.
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Antenna lengths

- **Dipole length:** $L \text{ (ft)} = 468 / F \text{ (MHz)}$
 - **$\frac{1}{4}\lambda$ vertical:** $L \text{ (ft)} = 234 / F \text{ (MHz)}$
 - L is length in feet, F is frequency in MHz.
 - Divide a full-wave loop by **4** for one side of a **Quad** loop.
 - Divide a full-wave loop by **3** for one side of a **Delta** loop.
 - A 3-element Yagi has an exam-pool gain figure of **9.7 dBi**.
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General Class HF band allocations

Mode segments by band for General-class privileges. Bolded edges appear in the question pool; the ARRL band plans carry more detail, but this is limited to the segments relevant to the General exam.

Band	CW / data segment	Phone / voice segment	Notes
160 m	1.800–2.000 MHz (all modes)	—	All modes band-wide
80 m	3.525–3.600 MHz (data ~3.570–3.600)	3.800–4.000 MHz	Often split: 80 m CW / 75 m voice
60 m	4 USB channels + 5351.5–5366.5 kHz segment	(USB channels)	See 60 m table below
40 m	7.025–7.125 MHz	7.175–7.300 MHz	
30 m	10.100–10.150 MHz	—	CW & data only — 200 W max , no phone, no image
20 m	14.025–14.150 MHz (RTTY/PSK31 focus 14.070–14.100)	14.225–14.350 MHz	
17 m	18.068–18.110 MHz	18.110–18.168 MHz	
15 m	21.025–21.200 MHz	21.275–21.450 MHz	
12 m	24.890–24.930 MHz	24.930–24.990 MHz	
10 m	28.000–28.300 MHz	28.300–29.700 MHz	29.0–29.7 commonly FM/repeaters

General Class power: up to **1500 W PEP**, except **200 W max on 30 m** and **100 W for propagation beacons**.

Notes:

- The 80 m band is often broken into the 80 m CW portion and 75 m voice portion. This more accurately reflects the actual wavelengths used and differentiates voice from CW.
- On 20 m, the RTTY subband has PSK31 activity focused at 14.070 MHz.

60 m channel detail

The FCC references 60 m by **channel center frequency**; operators tune **USB 1.5 kHz below center** (dial). Center is canonical; dial is secondary.

#	Center frequency	USB dial frequency	Max power	Notes
1	5332.0 kHz	5330.5 kHz	100 W ERP	
2	5348.0 kHz	5346.5 kHz	100 W ERP	
3	5373.0 kHz	5371.5 kHz	100 W ERP	
4	5405.0 kHz	5403.5 kHz	100 W ERP	
seg	5351.5–5366.5 kHz	—	9.15 W ERP	All modes, 2.8 kHz max

General-class band summary table

Band	Frequency	General-class segments	Voice	Max power	Notes
160 m	1.8 MHz	1800–2000 kHz	LSB	1500 W PEP	Night owl
80 m	3.5 MHz	3525–3600 / 3800–4000 kHz	LSB	1500 W PEP	Evenings & nights
60 m	5 MHz	4 channels + segment	USB	100 W ERP / 9.15 W ERP	USB channels
40 m	7 MHz	7025–7125 / 7175–7300 kHz	LSB	1500 W PEP	Best days & evenings
30 m	10.1 MHz	10100–10150 kHz (CW & data)	—	200 W PEP	CW & data only
20 m	14 MHz	14025–14150 / 14225–14350 kHz	USB	1500 W PEP	Best days & nights
17 m	18 MHz	18068–18168 kHz	USB	1500 W PEP	Best days & evenings
15 m	21 MHz	21025–21200 / 21275–21450 kHz	USB	1500 W PEP	Best days
12 m	24 MHz	24890–24990 kHz	USB	1500 W PEP	Best days
10 m	28 MHz	28000–29700 kHz	USB	1500 W PEP	Best days
6 m	50 MHz	50–54 MHz	USB	1500 W PEP	
2 m	144 MHz	144–148 MHz	USB	1500 W PEP	
1.25 m	222 MHz	222–225 MHz	USB	1500 W PEP	
70 cm	420 MHz	420–450 MHz	USB	1500 W PEP	
33 cm	902 MHz	902–928 MHz	USB	1500 W PEP	
23 cm	1240 MHz	1240–1300 MHz	USB	1500 W PEP	

HF data: 2.8 kHz maximum bandwidth.

Technician Study Sheet

The following pages cover Technician-class material from the same study set: information extracted from the question pool that rewards memorization. Practice tests are available online (hamstudy.org is the current, all-class option; qrz.com also hosts practice tests).

Technician class frequency privileges (1500 W PEP max)

Band	Frequency	Notes
6 meters	50–54 MHz	
2 meters	144–148 MHz	
1.25 meters	222–225 MHz	219–220 MHz point-to-point digital links
70 centimeters	420–450 MHz	435–438 MHz satellite subband; 446.000 MHz FM calling
23 centimeters	1240–1300 MHz	

Wavelength (meters) = 300 / Frequency (MHz). Bands are organized with narrow modes near the bottom edge.

- **Valid US callsigns** start with A, K, N, or W (WANK) and contain 1 digit (0–9).
- The **FCC** regulates radio use in the US; the **ITU** organizes radio operations internationally.
- **License period:** 10 years, with a 2-year grace period; can renew 90 days before expiration.
- Know **Local**, **Remote**, and **Automatic** control of a transmitter.
- A **club** must have at least 4 members to be eligible for a club license.
- **Identification:** must ID every 10 minutes and at the end of a transmission — CW under 20 WPM is OK.

Good amateur practices: listen before transmitting; to break into a conversation, just transmit your callsign; use minimum power. No broadcasting, no music, no codes/ciphers intended to obscure a message, no swearing, no ethnic/racial slurs, no business.

Filters: spurious emissions / harmonics (your problem) require a low-pass filter at the transmitter. RF front-end / fundamental overload (their problem) requires a high-pass filter at the TV receiver.

Repeater frequency separation (split): 2 m band = 600 kHz; 70 cm band = 5.0 MHz.

Bandwidth:

Signal	Bandwidth
CW	150 Hz
SSB	2–3 kHz
FM	5–15 kHz
TV (NTSC)	6 MHz

Ohm's law and power formulas (Technician)

Cover the value you need and divide or multiply the remaining values:

- $E = I \times R$; $I = E / R$; $R = E / I$
- $P = I \times E$; $I = P / E$; $E = P / I$

Unit conversions:

Prefix	Power	Value	Prefix	Power	Value
Mega	10^6	1,000,000	Micro	10^{-6}	0.000001
Kilo	10^3	1,000	Milli	10^{-3}	0.001
			Pico	10^{-12}	0.000000000001
			Nano	10^{-9}	0.000000001

- **Voice frequencies** are 300–3000 Hz; **radio waves** are above 20,000 Hz.

Measuring units: Hertz (frequency); Ampere (current); Volts (electromotive force); Ohm (resistance); Watt (power).

Radio modes (Technician)

- Amplitude Modulation (AM), Single Sideband (SSB), and Code (CW) are all related.
- A **product detector** is used for **SSB/CW** reception; a **discriminator** is used for **FM**.
- FM antenna polarization is vertical by convention (best for cars); SSB/CW is horizontal.
- Upper sideband (USB) is used above about 10 MHz.
- Continuous Tone Coded Squelch System (CTCSS) uses a subaudible tone to unmute the receiver.

Q-signals (Technician):

Signal	Meaning	Signal	Meaning
QRM	Man-made noise	QRN	Atmospheric noise
QSY	Change frequency	QTH	Current / home location

- CQ means calling any station. Respond with the station callsign (this is) your callsign.

Emergency communications: RACES only serves government agencies; ARES serves government and NGO groups. Messages contain the name of the sender in the preamble, no more than 25 words, and a check.

- A **3 dB** change is double; **6 dB** is quadruple; **10 dB** is a ten-fold increase.
- **Inductors** store energy in a **magnetic** field; **capacitors** store energy in an **electric** field.
- A **bipolar transistor** has an emitter; a **field-effect transistor** has a gate.
- **SWR** of 1:1 is perfect; near 2:1 transmitters generally start reducing power output.

Antenna types: half-wave dipole; quarter-wave vertical. Wavelength (m) = 300 / Frequency (MHz).

Technician operating reference

Identification: all operators must identify with their FCC callsign every 10 minutes while talking and at the end of a transmission, or it is an illegal "unidentified transmission." There is no requirement to ID at the beginning. CW (Morse) at less than 20 words per minute is always an acceptable way to ID.

License period: an amateur license is valid for 10 years. It can be renewed no earlier than 90 days in advance. There is a two-year renewal grace period during which the license is not valid but can be renewed with no loss of privileges (no retest). Transmitting during the grace period is not allowed.

SWR: Standing Wave Ratio measures how well the antenna is matched to the radio. A 1:1 SWR means all radio output power is transferred to the antenna and is ideal. A high SWR can overheat the radio, so most radios have circuitry that lowers output levels as SWR increases.

Radio functions:

- **VFO** (Variable Frequency Oscillator) knob: changes operating frequency.
- **RIT** (Receiver Incremental Tune): fine-tunes the receiver without affecting the transmit frequency.
- **Squelch:** keeps the radio quiet (muted) when there is no signal present.

Frequency bands (Technician quick reference)

Band	Frequency	Wavelength
HF	3–30 MHz	100–10 m
VHF	30–300 MHz	10 m–1 m
UHF	300–3000 MHz	1 m–10 cm

Technician band allocations

Band	Frequency	Technician segments	Notes
23 cm	1240–1300 MHz	All modes	
70 cm	420–450 MHz	All modes	"Not available north of Line A"; satellite 435–438; 5 MHz repeater split
1.25 m	219–225 MHz	All modes	219–220 MHz point-to-point digital links
2 m	144–148 MHz	CW 144.0–144.1; all modes above	600 kHz repeater split
6 m	50–54 MHz	CW 50.0–50.1; all modes above	
10 m	28.000–29.700 MHz	CW & data 28.000–28.300; phone 28.300–28.500	HF privileges informational only — not on the Technician test
15 m	21.025–21.450 MHz	CW 21.025–21.200	HF — informational
40 m	7.025–7.300 MHz	CW 7.025–7.150	HF — informational
80 m	3.500–4.000 MHz	CW 3.525–3.600	HF — informational

Notes:

- No-Code Technicians have all privileges on the 6 m, 2 m, 1.25 m, 70 cm, and 23 cm bands.
- The HF (80 m–10 m) privileges shown are informational only — they are not on the Technician test.
- Technician licensees may use up to **1500 W PEP** on the VHF and higher bands, and are **limited to 200 W on the HF bands**.